

5	(a) the <u>total</u> momentum of a system (of colliding particles) remains constant	M1	
	provided there is no resultant external force acting on the system/isolated or closed system	A1	[2]
	(b) (i) the <u>total</u> kinetic energy before (the collision) is equal to the total kinetic energy after (the collision)	B1	[1]
	(ii) $p (= mv = 1.67 \times 10^{-27} \times 500) = 8.4 (8.35) \times 10^{-25} \text{ N s}$	A1	[1]
	(iii) 1. $mv_A \cos 60^\circ + mv_B \cos 30^\circ$ or $m(v_A^2 + v_B^2)^{1/2}$	B1	
	2. $mv_A \sin 60^\circ + mv_B \sin 30^\circ$	B1	[2]
	(iv) 8.35×10^{-25} or $500m = mv_A \cos 60^\circ + mv_B \cos 30^\circ$ and $0 = mv_A \sin 60^\circ + mv_B \sin 30^\circ$ or using a vector triangle	C1	
	$v_A = 250 \text{ m s}^{-1}$	A1	
	$v_B = 430 (433) \text{ m s}^{-1}$	A1	[3]

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